

Case Study: MOM Compliance Assistant

Building Guardrails for Real-World Labor Laws

The Goal

The mission was simple but high-stakes: build a specialized AI agent to help workers navigate **Singapore Ministry of Manpower (MOM)** employment practices without the "fluff" or the risk of bad advice. The focus was on 100% accuracy and zero hallucinations, specifically regarding work hours and overtime.

1. The Reality Check

Standard AI models often try too hard to be helpful. They might guess legal details or mix up labor laws from different countries. For someone working **12-hour rotating shifts**, a small mistake in explaining overtime (OT) caps can lead to real-world headaches. This project was designed to "jail" the AI within official documentation to ensure every answer was backed by the source.

2. The Strategy

- **Walled Garden:** I pinned the agent's knowledge strictly to the official MOM Employment Practices URL.
- **The "Kebab" Guardrail:** I built in a hard-stop instruction. If a user asks something irrelevant—like where to find the best kebabs—the agent politely declines instead of guessing.
- **Handling the Nuance:** I engineered the prompt to grasp complex rules, like the **3-week averaging** regulation for shift workers, which most general AI models usually miss.

3. Does it Actually Work?

- **Staying on Task:** The agent passed the "Out-of-Scope" test by refusing to give food recommendations, proving the guardrails hold.
- **Pinpoint Accuracy:** It successfully explained that for 12-hour shifts, OT only kicks in when the *average* weekly hours exceed 44 over a 3-week stretch.
- **No Walls of Text:** I capped responses at 120 words to keep things fast, scannable, and professional.

4. The Takeaway

By trading "creative freedom" for strict **constraint logic**, I built a tool that actually works for people in high-stakes environments. It's a utility-first approach: reliable, legally aligned, and built for the Singaporean workforce.